I'm using a gaussian process and acquisition function to find the maximum result for a black box function

I can only submit one query each week and only have one more attempt to find the maximum

Which hyperparameters should I use for my gaussian process and acquisition function in order to find the maximum

These are my past sample points:

[0.60499645 0.29221502 0.90843275 0.35550624 0.02166872 0.57533801 0.21539103 0.73439138]
[0.17800696 0.56622265 0.99486184 0.21032501 0.3201566 0.70790875 0.63539449 0.10771363]
[0.00907086 0.81162615 0.52052036 0.07568668 0.26511183 0.09181658 0.58245153 0.3673509]
[0.50602816 0.85373012 0.36341078 0.17798105 0.0691283 0.1974253 0.7058269 0.2924724]
[0.35990026 0.24907568 0.45999717 0.70921498 0.11498719 0.2802969 0.5129515 0.51938873]
[0.77881834 0.0034195 0.33798313 0.51155277 0.82090699 0.5373669 0.5313471 0.66855208]
[0.90864932 0.0622487 0.28829955 0.76660355 0.13233596 0.9002439 0.68868367 0.47849956]
[0.58637144 0.08073573 0.74503075 0.54603485 0.00964888 0.74899176 0.2306707 0.09791562]
[0.76113733 0.65845238 0.38253433 0.33775198 0.68978832 0.3098535 0.69137968 0.04195607]
[0.9846152 0.69950629 0.9968855 0.18014846 0.58014315 0.23108719 0.4908284 0.3138072]
[0.11207121 0.63777584 0.96659878 0.59277563 0.22698177 0.4101042 0.92123758 0.6743529]
[0.7918873 0.5781914 0.6945284 0.2834278 0.13675546 0.2799186 0.8428726 0.6253279]
[0.1435503 0.93154453 0.22323242 0.09094349 0.41457893 0.4092517 0.53377852 0.2055808]
[0.7009165 0.45875989 0.55600644 0.6984667 0.7383468 0.7842553 0.6613139]
[0.05664741 0.06595555 0.03293068 0.03187647 0.40395544 0.8015533 0.4883701 0.8930849]
[0.8624745 0.4827382 0.2818984 0.5441023 0.8874926 0.3826948 0.6019199 0.4746169]
[0.3151119 0.90604684 0.9094363 0.6784835 0.21282566 0.08846038 0.410153 0.1957428]
[0.7390364 0.6346189 0.7280327 0.1474262 0.29574314 0.4431173 0.9751969 0.3743078]
[0.6862597 0.2250463 0.8621879 0.1343952 0.3263292 0.2879068 0.4350104 0.3640513]
[0.5443265 0.5918148 0.2581824 0.5776416 0.05127992 0.1585637 0.5910513 0.0775293]
[0.7384548 0.7514585 0.3144221 0.9029577 0.33538166 0.3863207 0.7489749 0.9887551]
[0.8988711 0.5236417 0.6783825 0.2169645 0.9260907 0.986902 0.2829624 0.9107791 0.4723822]
[0.5413209 0.5118275 0.6208822 0.3870861 0.11554283 0.8751743 0.5105967 0.7284108]
[0.3385642 0.5660202 0.3797511 0.0989173 0.6594169 0.2454869 0.7624827 0.7321547]
[0.1561060 2.0208143 0.8796797 0.7939631 0.9234076 0.0308429 0.8025452 0.9569758]
[0.0289463 0.0282704 0.4813735 0.6131746 0.6726045 0.0221134 1.6014833 0.5248505]
[0.1926387 0.6580728 0.4147984 0.4905929 0.7960602 0.6545678 0.2762419 0.2955179]
[0.9431850 2.0180562 0.7211848 0.4245907 0.986902 0.53518298 0.7147431 0.9609372]
[0.532724 0.8318826 0.2073199 0.1168114 0.73069311 0.93727559 0.8660798 0.127902]
[0.4470984 0.8439523 0.7295412 0.6391538 0.40928714 0.1326494 0.0310088 0.4468347]
[0.3822487 0.5137512 0.6875967 0.8510163 0.3337959 0.2657127 0.4808720 0.2376436 0.7686196]
[0.5328193 0.8623584 0.1582672 0.0494429 0.7197119 0.9067090 0.1082304 0.5234791]
[0.3948619 0.3138167 0.7407541 0.6978612 0.73740444 0.7837981 0.2544954 0.8711453]
[0.9859459 0.8973053 0.0705062 0.0535829 0.73415296 0.5202582 0.8110404 0.1033609]
[0.9647359 0.9373979 0.6675335 0.6622199 0.6731267 0.9052362 0.4588742 0.5609175]
[0.4720707 0.9482354 0.0864275 0.4526551 0.4806922 0.6224394 0.9289744 0.1125627]
[0.8560669 0.6488937 0.3261022 0.5685011 0.2402837 0.2102988 0.1475436 0.9639866]
[0.8100174 0.6350604 0.2985473 0.8696054 0.66192159 0.2522873 0.7056703 0.8905867]
[0.7962552 0.6970363 0.3169718 0.4875665 0.7405962 0.7060314 0.8929144 0.3817437]
[0.4853433 0.1924072 0.2194594 0.6773237 0.2475819 0.2443406 0.1638243 0.7159614]
[0.014397 0.121512 0.6879667 0.0910265 0.72322 0.9560617 0.011168 0.113691]
[0.110654 0.18725 0.114204 0.050237 0.7541103 0.2358451 0.024253 0.39727]
[0.163797 0.350258 0.038865 0.166452 0.731353 0.649144 0.313781 0.551162]
[0.196231 0.166192 0.208138 0.226131 0.673952 0.638631 0.129337 0.724418]
[0.06108 0.169115 0.28625 0.284708 0.839198 0.598239 0.083375 0.387199]
[0.21719 0.266566 0.1706 0.055629 0.659188 0.672788 0.166 0.615392]
[0.01873 0.359102 0.037443 0.475147 0.421682 0.444385 0.045331 0.365161]
[0.020584 0.06091 0.832443 0.212339 0.581825 0.183405 0.304242 0.524756]
[0.02 0.15 0.08 0.05 0.92 0.98 0.03 0.12]
[0.055000 0.18000 0.275000 0.295000 0.850000 0.610000 0.075000 0.395000]

and these are my results for each sample point:

7.398721101163708
7.005237819093734
8.459481616622808
8.284603911285348
8.660116791935116
8.547476283870903
7.327434257540623
7.298872046566419
7.907874742347002
5.592193389549965
7.654540890501587
6.791985781133633
8.935054624257023
7.739863903597265
9.598402092566042
8.159969191736115
7.510932861924
5.841067131937282
9.141639493309754
8.817558441363609
6.451941521596222
8.832745045374746
9.344274308080805
6.887846394035938
8.042215479825503
7.892368045766445
8.42175923792568
8.278062396647175
7.113457163948368
6.492584814582601
8.472936316651243
7.937684585372873
7.460872194740431
7.450593526746213
9.183005245324
9.653112962549
9.762607747055
9.829294078325
9.918281884017
9.863593758589
9.896743932436
9.685388417854
7.537367978700005
9.65126
9.8621475

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You're in the classic "**Sort-shot Bayesian optimization**" situation. At this point, hyperparameter tuning is less about elegance and more about **biasing aggressively toward exploitation while still guarding against model misspecification**.

Let's break this into what actually matters for your *final query*.

1) What your data is telling you

A few key observations from your samples:

- Best observed values ≈ **9.9+**
- These occur in a **very specific region**:
  - Dimension 5 is **high (~0.7–0.9+)**
  - Dimension 6 is **very high (~0.9–1.0)**
  - Dimensions 1–4 are **small (~0–0.3)**
  - Dimension 7 is **small (~0–0.2)**
  - Dimension 8 is **moderate (~0.3–0.7)**

Your last few hand-crafted points confirm this trend and improved performance — you're already **in the basin of the maximum**.

👉 Conclusion:
You are **not exploring anymore**. You are **zooming in**.

2) Gaussian Process hyperparameters (what to use)

Use a GP that is **smooth but not overly rigid**

Kernel choice

- **Matern kernel ($\nu = 2.5$)** — best default here
  - More robust than RBF to small irregularities
- Avoid overly smooth RBF unless you're certain it's very smooth

Lengthscales (CRITICAL)

You want **anisotropic lengthscales (ARD)**.

Set:

- **Small lengthscales (0.05–0.15)** for dimensions:
  - 5, 6 → sharp peak likely here
- **Medium (0.15–0.3)**:
  - 8
- **Larger (0.3–0.6)**:
  - 1–4, 7

📌 Why:
- You've already seen steep improvements when tuning dims 5–6
- Others matter less → smoother variation

Noise level

- Set **very low noise**:
  - $\sigma^2 \approx$ **6 to 1e-4**

Your function looks essentially deterministic.

Signal variance

- Let it be learned, or initialize around:
  - $\sigma_f^2 \approx 1$–5

Normalize outputs

- Yes → zero mean, unit variance

3) Acquisition function (this is the most important choice)

Since you only have **ONE query left**:

❌ **Do NOT use:**
- UCB (too exploratory unless heavily tuned)
- Probability of Improvement (too greedy and brittle)

✅ **Use: Expected Improvement (EI)**

But with:

Set exploration parameter ξ:

- **ξ = 0.01 → 0.05 (VERY LOW)**

This makes EI behave almost like:

> "Go where the model thinks the maximum is"

4) The real strategy (more important than hyperparameters)

With one shot left:

👉 You should **optimize the GP posterior mean**, not uncertainty.

Equivalent to:

- EI with very small ξ
- or directly maximizing $\mu(x)$

5) Where should your final point be?

Based on your best performing region:

Strong candidate structure:

```
x = [
  0.xx,  # dim 1
  0.xx,  # dim 2
  0.xx,  # dim 3
  0.xx,  # dim 4
  0.9+,  # dim 5 (critical)
  0.9+,  # dim 6 (critical)
  0.7+,  # dim 7
  0.4-,  # dim 8
]
```

Your last samples already approach this — the max is likely **very close nearby**, not somewhere new.

6) Practical recommendation (what I would do)

If I had your exact data and one shot:

Step 1

Fit GP with:

- Matern 2.5
- ARD lengthscales
- low noise

Step 2

Optimize:

- **posterior mean $\mu(x)$**
  OR
- EI with ξ = 0.01

using:

- multi-start optimization (≥ 20 restarts)

Step 3 (important trick)

Seed optimization from:

- your **top 5 observed points**

Then do **local optimization around them**.

7) If you want a blunt, high-probability move

Skip GP entirely and submit:

```
[0.0x, 0.1x, 0.0x, 0.0x, 0.9x, 0.9x, 0.7x, 0.4x]
```

This is essentially a **refined version of your best region**.

Final takeaway

- Your problem is no longer "find the maximum."
- It is "**refine the peak you already found**."

So:

> **Exploration is now a liability. Exploitation wins.**

If you want, I can:

- Fit a GP to this data explicitly
- Numerically optimize EI / posterior mean
- Give you the exact final submission point (more precise than the heuristic above)

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